

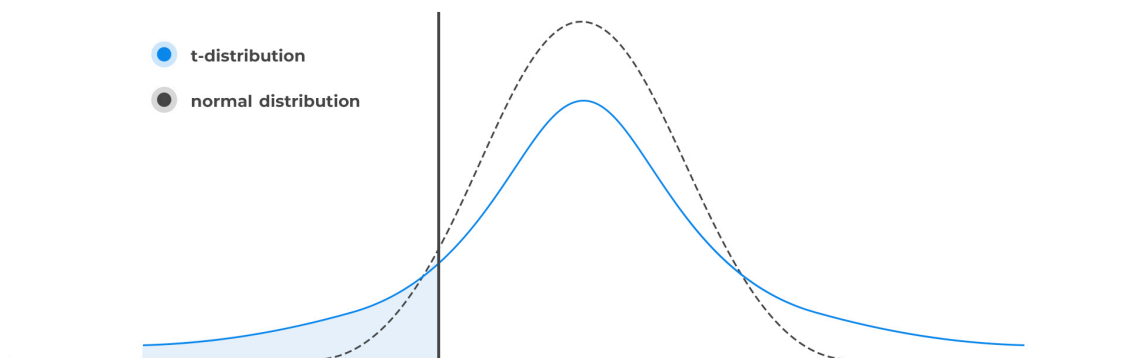
t-distribution

It is a probability distribution used to estimate population parameters when sample sizes are small ($n < 30$) or when the population standard deviation is unknown.

It is bell-shaped and symmetric, like the normal distribution, but has heavier tails, making it more robust for smaller datasets.



T-distribution vs Normal Distribution



Characteristic

- shape: similar to the normal distribution (z), but with wider tails to account for greater uncertainty.
- Degree of freedom (df): the shape depends on df , $n-1$
- Purpose: used for calculating the confidence interval and hypothesis test (t -test) when the sample size is small
- Mean: Centered at zero, similar to the standard normal distribution
- We use it when:
 - The sample size is small ($n < 30$)
 - the population standard deviation (σ) is unknown, requiring the use of the sample standard deviation (s)
 - The data follows an approximately normal distribution